Design Verification Plan and Report — T8-40g

VBF-T8R1-DVPR-01 | Case t8_r1 — Long-endurance drone battery | 2026-08-25

Prepared: ________ Reviewed: ________ Approved: ________

Design Verification Plan and Report (virtual test version) — T8-40g

- **Document code**: VBF-T8R1-DVPR-01
- **Case**: t8_r1 — long-endurance drone battery
- **Generation date**: 2026-08-25
- **Signature**: Prepared: ________ Reviewed: ________ Approved: ________

All result values are mechanically taken from tool output files under `cell/`. Determinatios compare against entry-0 criteria (energy density ≥ 446.18 Wh/kg, 5C retention ≥ 90%, mass ≤ 40 g) and protocol safety defaults (T_max ≤ 333.15 K, no plating).

Verification Items

#	Item	Condition	Result value	Determination	Source
1	1C discharge capacity	`run-pyamm --protocol 1C_discharge` (DFN, lumped thermal)	6.9427 Ah (T_max 299.76 K)	PASS (capacity consistent with nominal 6.1494 Ah)	`cell/r5_final_1c_dfn.json`
2	Energy density	`calc-energy` on 1C discharge, contract mass caliber	634.32 Wh/kg (1027.10 Wh/L)	PASS ≥ 446.18 Wh/kg	`cell/r5_final_energy.json`
3	Cell mass	contract formula Σ layer ×(1−ε)×density×area	39.798 g	PASS ≤ 40 g	`cell/r5_final_energy.json`
4	5C discharge capacity retention	`run-pyamm --protocol 5C_discharge` ÷ same-params 1C capacity	6.8571 Ah / 98.77% (T_max 319.99 K)	PASS ≥ 90%	`cell/r5_final_5c_dfn.json`, `cell/r5_final_retention.json`
5	4C fast-charge temperature rise	`run-pyamm --protocol 4C_charge_45C --thermal lumped`	T_max 330.41 K	PASS ≤ 333.15 K (margin 2.74 K)	`cell/r5_final_4c45_dfn.json`
6	4C fast-charge lithium plating	same protocol + `--plating`; negative electrode potential < 0 V	anode potential min +0.0082 V	PASS (no plating)	`cell/r5_final_4c45_dfn.json`
7	Voltage window	parameter set limits	2.5 – 4.2 V	PASS (nominal window used by all protocols)	`cell/chen2020_base_probe.json`
8	Charge acceptance at 4C	`4C_charge_45C` capacity integral	0.9395 Ah before 4.2 V cut-off	INFORMATIONAL — voltage-limited at 4C; not a task criterion	`cell/r5_final_4c45_dfn.json`

Items Marked N/A

Item	Status
Nail penetration	N/A (beyond pure simulation boundary, requires physical experiment)
Overcharge to thermal runaway	N/A (beyond pure simulation boundary, requires physical experiment)
Crush	N/A (beyond pure simulation boundary, requires physical experiment)
Drop	N/A (beyond pure simulation boundary, requires physical experiment)
Cycle life	N/A (aging model exists in Chen2020 base but was not exercised — no cycle-life criterion in task)
Rate-pulse internal resistance	N/A (no pulse protocol in task criteria; DC-IR 0.684 mΩ mechanically derived in `r5_final_energy.json` as informational)

Conclusion

- **Summary**: 7/7 simulated verification items PASS against entry-0 criteria and protocol safety defaults. Final candidate T8-40g: ED 634.32 Wh/kg, 5C retention 98.77%, mass 39.798 g, 4C-45 °C T_max 330.41 K, no plating.
- **Uncovered items (paper limitations)**: nail penetration, overcharge-to-thermal-runaway, crush, drop, cycle life, rate-pulse internal resistance — all N/A beyond pure simulation boundary or outside task criteria.
- **Honest caveats**: 4C charge acceptance is voltage-limited to 0.94 Ah (item 8); safety thresholds are protocol defaults (task silent on safety); contract-mass caliber excludes electrolyte.